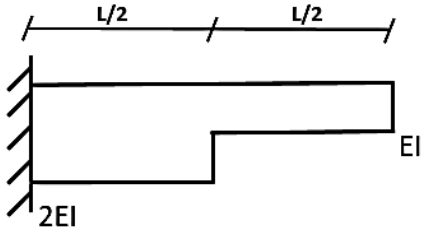
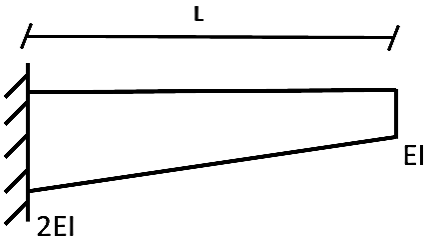
تمرین سری یک (روش های المان محدود)

(دانشجو: سبا آزادگان / شماره دانشجویی: ۶۰۲[redacted]۹۷)

1. **محاسبه پارت ۳ و ۴ برای المان تیر خمشی با اثر برش انجام شود.**
2. **ماتریس سختی سازه رو به رو محاسبه شود.**

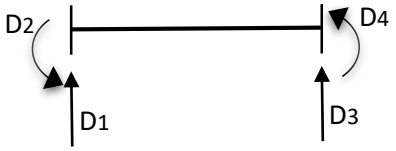


3. **اگر المان بالا به صورت tapper باشد، ماتریس سختی محاسبه شود.** (تشویقی)



4. **بر اساس چهار ویژگی بر پا نمودن توابع میدان، چگونه می توان یک ساختار پارامتری برای ۴ المان محدود LST , Q5 , Q8 , Q9 بدست آورد.**

المان تیر خمشی



کار کمینه

$$D = \int_0^L \frac{1}{EI}\, M_R(x) \left(\frac{\partial M_R}{\partial P_i}\right) dx + \int_0^L \frac{f_s}{GA}\, V_R(x) \left(\frac{\partial V_R}{\partial P_i}\right) dx$$

کار مجازی

$$D = \int_0^L \frac{1}{EI}\, M_R(x) M_V(x) dx + \int_0^L \frac{f_s}{GA}\, V_R(x) V_V(x) dx$$

حل:

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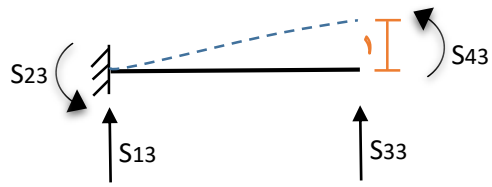


part 3

$M_R(x) = S_{33}(L-x) + S_{43}$	$V_R(x) = -S_{43}$
$M_V(x) = L - x$	$V_V(x) = -1$

$$D_1 = 1 = \frac{1}{EI}\int_0^L [S_{33}(L-x) + S_{43}](L-x)dx + \frac{f_s}{GA}\int_0^L (-S_{43})(-1)dx$$

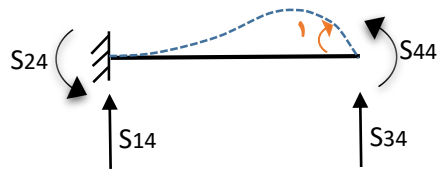
$$\mathbf{I)}\ \frac{\mathbf{1}}{\mathbf{EI}} * \frac{\mathbf{L^2(3S_{43} + 2LS_{33})}}{\mathbf{6}} + \frac{\mathbf{S_{43}L}}{\mathbf{GA}} = \mathbf{1}$$

$M_R(x) = S_{33}(L-x) + S_{43}$	$V_R(x) = -S_{43}$
$M_V(x) = 1$	$V_V(x) = 0$

$$D_2 = 0 = \frac{1}{EI}\int_0^L [S_{33}(L-x) + S_{43}](1)dx + \frac{f_s}{GA}\int_0^L (-S_{43})(0)dx$$

$$\mathbf{II)}\ \frac{\mathbf{1}}{\mathbf{EI}} * \frac{\mathbf{L(2S_{43} + LS_{33})}}{\mathbf{2}} = \mathbf{0}$$

$\boldsymbol{S_{33} = \frac{12GA * EI}{L^2(GAL - 6EI)}}$	$\boldsymbol{S_{13} = -S_{33} = -\frac{12GA * EI}{L^2(GAL - 6EI)}}$
$\boldsymbol{S_{43} = -\frac{6GA * EI}{L(GAL - 6EI)}}$	$\boldsymbol{S_{23} = S_{33}L + S_{43} = \frac{18GA * EI}{L(GAL - 6EI)}}$



part 4

$M_R(x) = S_{34}(L-x) + S_{44}$	$V_R(x) = -S_{34}$
$M_V(x) = L - x$	$V_V(x) = -1$

$$D_1 = 0 = \frac{1}{EI}\int_0^L [S_{34}(L-x) + S_{44}](L-x)dx + \frac{f_s}{GA}\int_0^L (-S_{34})(-1)dx$$

$$\boldsymbol{I)\ \frac{1}{EI} * \frac{L^2(3S_{44} + 2LS_{34})}{6} + \frac{S_{34}L}{GA} = 0}$$

$M_R(x) = S_{34}(L-x) + S_{44}$	$V_R(x) = -S_{34}$
$M_V(x) = 1$	$V_V(x) = 0$

$$D_2 = 1 = \frac{1}{EI}\int_0^L [S_{34}(L-x) + S_{44}](1)dx + \frac{f_s}{GA}\int_0^L (-S_{34})(0)dx$$

$$\boldsymbol{II)\ \frac{1}{EI} * \frac{L(2S_{44} + S_{34}L)}{2} = 1}$$

$S_{34} = -\dfrac{6AEGI}{AGL^2 + 12EI}$	$S_{14} = -S_{34} = \dfrac{6AEGI}{AGL^2 + 12EI}$
$S_{44} = \dfrac{4EI * (AGL^2 + 3EI)}{L * (AGL^2 + 12EI)}$	$S_{24} = S_{34}L + S_{44} = \dfrac{4\mathrm{EI} + 6\mathrm{EI} * \mathrm{AG}L^2}{L(AG * L^2 + 12EI)}$

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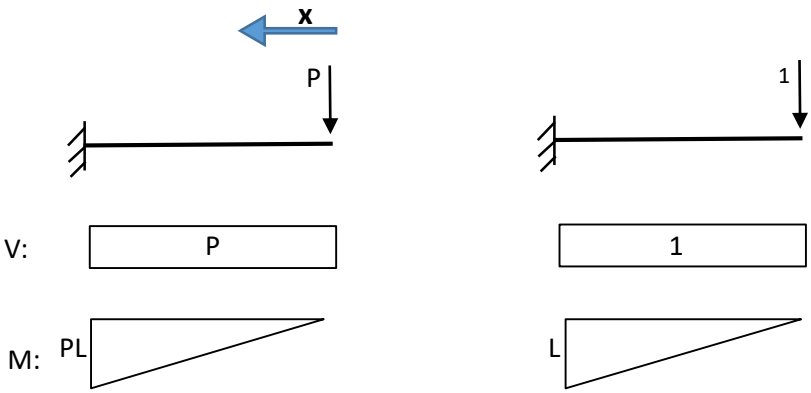
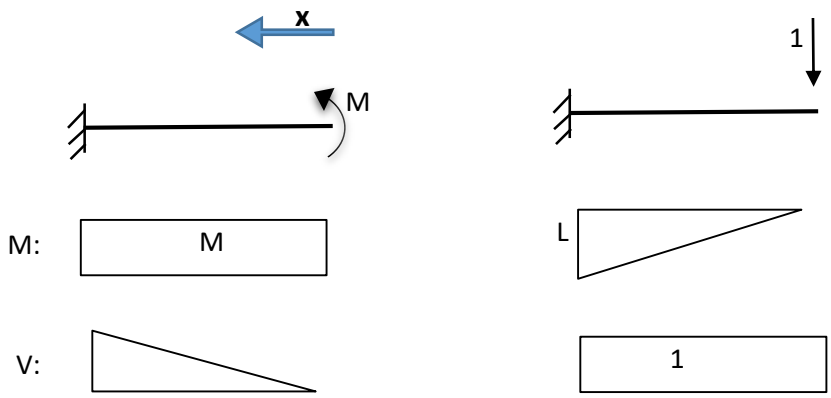


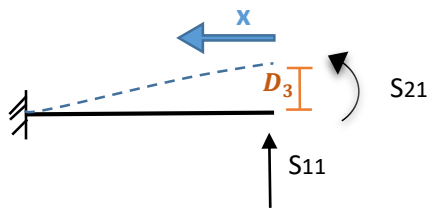
$$D_3 = \left[\frac{1}{EI}\int_0^{L/2} Px * x\,dx + \frac{1}{2EI}\int_{L/2}^{L} Px * x\,dx\right] + \frac{f_s}{GA}\int_0^{L} P * 1\,dx$$

$$D_3 = \frac{9L^3}{48EI} + \frac{L}{GA}$$



$$D_4 = \left[\frac{1}{EI}\int_0^{L/2} M * x\,dx + \frac{1}{2EI}\int_{L/2}^{L} M * x\,dx\right] + \frac{f_s}{GA}\int_0^{L} \frac{M}{L} * x * 1\,dx$$

$$D_4 = \frac{5L^2}{16EI} + \frac{L}{2GA}$$



$M_R(x) = S_{11}x + S_{21}$	$V_R(x) = -S_{11}$
$M_V(x) = x$	$V_V(x) = -1$

$$D_1 = \frac{L}{GA} = \int_0^{L/2} \frac{1}{EI}\,(S_{11}x + S_{21})(x)dx + \int_{L/2}^{L} \frac{1}{2EI}\,(S_{11}x + S_{21})(x)dx + \int_0^L \frac{f_s}{GA}\,(-S_{11})(-1)dx$$

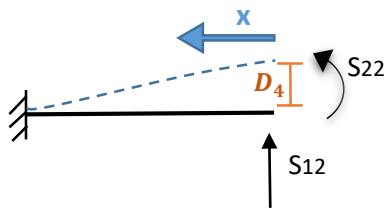
$$\boldsymbol{I)\ \frac{L^2(3S_{21} + S_{11}L)}{24EI} + \frac{L^2(9S_{21} + 7S_{11}L)}{48EI} + \frac{S_{11}L}{GA} = \frac{L}{GA}}$$

$M_R(x) = S_{11}x + S_{21}$	$V_R(x) = -S_{11}$
$M_V(x) = 1$	$V_V(x) = 0$

$$D_2 = \frac{9L^3}{48EI} = \int_0^{L/2} \frac{1}{EI}\,(S_{11}x + S_{21})(1)dx + \int_{L/2}^{L} \frac{1}{2EI}\,(S_{11}x + S_{21})(1)dx + \int_0^L \frac{f_s}{GA}\,(-S_{11})(0)dx$$

$$\boldsymbol{II)\ \frac{L(4S_{21} + S_{11}L)}{8EI} + \frac{L(4S_{21} + 3S_{11}L)}{16EI} = \frac{9L^3}{48EI}}$$

$S_{11} = \dfrac{3(-5AGL^3 + 64EI)}{11AGL^2 + 192EI}$	$S_{21} = \dfrac{L * (9AGL^3 + 48EIL - 80EI)}{11AGL^2 + 192EI}$



$M_R(x) = S_{12}x + S_{22}$	$V_R(x) = -S_{12}$
$M_V(x) = x$	$V_V(x) = -1$

$$D_1 = \frac{L}{2GA} = \int_0^{L/2} \frac{1}{EI}\,(S_{12}x + S_{22})(x)dx + \int_{L/2}^{L} \frac{1}{2EI}\,(S_{12}x + S_{22})(x)dx + \int_0^{L} \frac{f_s}{GA}\,(-S_{12})(-1)dx$$

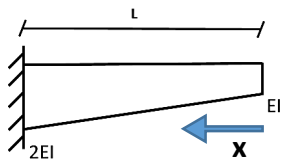
$$\boldsymbol{I)\ \frac{L^2(3S_{22} + S_{12}\,L)}{24EI} + \frac{L^2(9S_{22}\ +\ 7S_{12}L)}{48EI} + \frac{S_{12}L}{GA} = \frac{L}{2GA}}$$

$M_R(x) = S_{21}x + S_{22}$	$V_R(x) = -S_{21}$
$M_V(x) = 1$	$V_V(x) = 0$

$$D_2 = \frac{5L^2}{16EI} = \int_0^{L/2} \frac{1}{EI}\,(S_{12}x + S_{22})(1)dx + \int_{L/2}^{L} \frac{1}{2EI}\,(S_{12}x + S_{22})(1)dx + \int_0^{L} \frac{f_s}{GA}\,(-S_{12})(0)dx$$

$$\boldsymbol{II)\ \frac{L(4S_{22} + S_{12}\,L)}{8EI} + \frac{L(4S_{22}\ +\ 3S_{12}L)}{16EI} = \frac{5L^2}{16EI}}$$

$S_{12} = \dfrac{-\,25AGL^2 +\ 96EI}{11AGL^2 +\ 192EI}$	$S_{22} = \dfrac{5L(3AGL^2 +\ 8EI)}{11AGL^2 +\ 192EI}$



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$$\boldsymbol{I}_{(x)} = EI\left(\frac{x}{L} - 1\right) + 2EI = EI\frac{x}{L} + EI = \boldsymbol{EI}(\frac{\boldsymbol{x}}{\boldsymbol{L}} + \mathbf{1})$$

$$D_3 = \int_0^L \frac{1}{EI\left(\frac{x}{L} + 1\right)} Px * x\, dx + \frac{f_s}{GA}\int_0^L P * 1\, dx$$

$$D_3 = \frac{-0.2L^3}{EI} + \frac{L}{GA}$$

$$D_4 = \int_0^L \frac{1}{EI\left(\frac{x}{L} + 1\right)} M * x\, dx + \frac{f_s}{GA}\int_0^L \frac{M}{L} * x * 1\, dx$$

$$D_4 = \frac{0.7L^2}{EI} + \frac{L}{2GA}$$

$$D_1 = \frac{L}{GA} = \int_0^L \frac{1}{EI\left(\frac{x}{L} + 1\right)} (S_{11}x + S_{21})(x)dx + \int_0^L \frac{f_s}{GA}(-S_{11})(-1)dx$$

$$\boxed{I)\ \frac{-0.2S_{11}L^3}{EI} + \frac{0.7S_{21}L^2}{EI} + \frac{S_{11}L}{AG} = \frac{L}{GA}}$$

$$D_2 = \frac{-0.2L^3}{EI} = \int_0^L \frac{1}{EI\left(\frac{x}{L} + 1\right)} (S_{11}x + S_{21})(1)dx + \int_0^L \frac{f_s}{GA}(-S_{11})(0)dx$$

$$\boxed{II)\ \frac{L(0.3S_{21} + LS_{11} - 0.3LS_{11})}{EI} = \frac{-0.2L^3}{EI}}$$

$S_{11} = \frac{2(7AGL^3 + 15EI)}{5(-11AGL^2 + 6EI)}$	$S_{21} = -\frac{2L(-2AGL^3 + 10EIL + 35EI)}{5(-11AGL^2 + 6EI)}$

$$D_1 = \frac{L}{2GA} = \int_0^L \frac{1}{EI\left(\frac{x}{L}+1\right)} (S_{12}x + S_{22})(x)dx + \int_0^L \frac{f_s}{GA} (-S_{12})(-1)dx$$

$$\boxed{I)\ \frac{-0.2S_{12}L^3}{EI} + \frac{0.7S_{22}L^2}{EI} + \frac{S_{12}L}{AG} = \frac{L}{2GA}}$$

$$D_2 = \frac{0.7L^2}{EI} = \int_0^L \frac{1}{EI\left(\frac{x}{L}+1\right)} (S_{12}x + S_{22})(1)dx + \int_0^L \frac{f_s}{GA} (-S_{12})(0)dx$$

$$\boxed{II)\ \frac{L(0.3S_{22} + LS_{12} - 0.3LS_{12})}{EI} = \frac{0.7L^2}{EI}}$$

$S_{12} = \frac{-49AGL^2 + 15EI}{5(-11AGL^2 + 6EI)}$	$S_{22} = \frac{7L(-2AGL^2 + 5EI)}{5(-11AGL^2 + 6EI)}$

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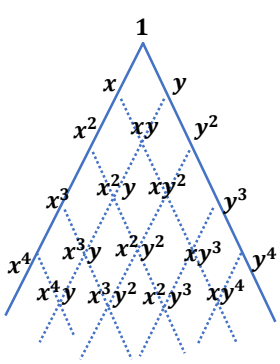
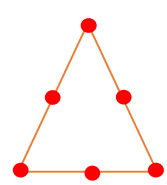


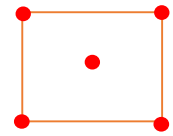


: LST

$$\begin{cases} u(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 xy + \alpha_6 y^2 \\ v(x,y) = \alpha_7 + \alpha_8 x + \alpha_9 y + \alpha_{10} x^2 + \alpha_{11} xy + \alpha_{12} y^2 \end{cases}$$

$$\varphi(x,y) = \beta_1 + \beta_2 x + \beta_3 y + \beta_4 x^2 + \beta_5 xy + \beta_6 y^2$$

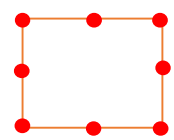
: Q5



$$\begin{cases} u(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 y^2 \\ v(x,y) = \alpha_8 + \alpha_9 x + \alpha_{10} y + \alpha_{11} x^2 + \alpha_{12} y^2 \end{cases}$$

$$\varphi(x,y) = \beta_1 + \beta_2 x + \beta_3 y + \beta_4 x^2 + \beta_5 y^2$$

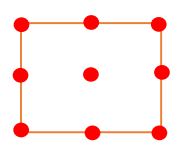
: Q8



$$\begin{cases} u(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 xy + \alpha_6 y^2 + \alpha_7 x^2 y + \alpha_8 xy^2 \\ v(x,y) = \alpha_9 + \alpha_{10} x + \alpha_{11} y + \alpha_{12} x^2 + \alpha_{13} xy + \alpha_{14} y^2 + \alpha_{15} x^2 y + \alpha_{16} xy^2 \end{cases}$$

$$\varphi(x,y) = \beta_1 + \beta_2 x + \beta_3 y + \beta_4 x^2 + \beta_5 xy + \beta_6 y^2 + \beta_7 x^2 y + \beta_8 xy^2$$

: Q9



$$\begin{cases} u(x,y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \alpha_4 x^2 + \alpha_5 xy + \alpha_6 y^2 + \alpha_7 x^2 y + \alpha_8 xy^2 + \alpha_9 x^2 y^2 \\ v(x,y) = \alpha_{10} + \alpha_{11} x + \alpha_{12} y + \alpha_{13} x^2 + \alpha_{14} xy + \alpha_{15} y^2 + \alpha_{16} x^2 y + \alpha_{17} xy^2 + \alpha_{18} x^2 y^2 \end{cases}$$

$$\varphi(x,y) = \beta_1 + \beta_2 x + \beta_3 y + \beta_4 x^2 + \beta_5 xy + \beta_6 y^2 + \beta_7 x^2 y + \beta_8 xy^2 + \beta_9 x^2 y^2$$